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Density of states

In order to know the actual number of electrons with a given energy one must know the number of states in a given system. Then multiplying the probability occupation we get the actual number of electrons. If $N(E)$ is the number of electrons in the system that have energy E and $Z(E)$ is the number of states

$$N(E)dE = Z(E)f(E)dE \quad \text{--- (1)}$$

Since we know that

$$E = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$$

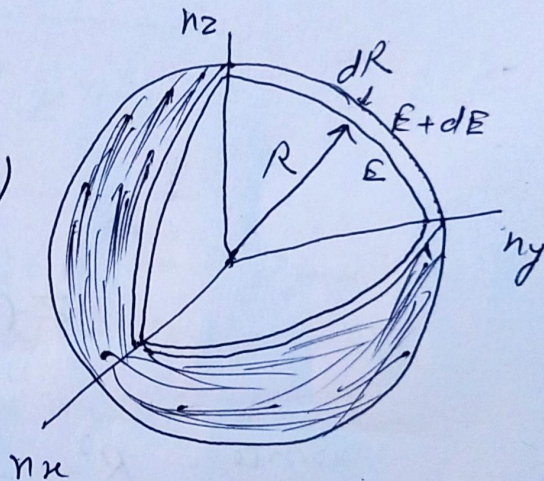
$$\text{Suppose } R^2 = n_x^2 + n_y^2 + n_z^2 \quad \text{--- (1)}$$

$$\& E_0 = \frac{\pi^2 \hbar^2}{8mL^2}$$

$$E_0 = \frac{\pi^2 \hbar^2}{8\pi^2 mL^2}$$

$$E_0 = \frac{\hbar^2}{8mL^2} \quad \text{--- (2)}$$

$$E = E_0 R^2 \quad \text{--- (2)}$$



Equation (1) represents the sphere of radius R , have energy E .

Thus the number of available states having energy E

$$= \frac{1}{8} \frac{4}{3} \pi R^3 \quad - (1)$$

Number of available states having energy $E+dE$, is

$$= \frac{1}{8} \frac{4}{3} \pi (R+dR)^3.$$

Thus the number of states having energy values lying between E & $E+dE$

$$\begin{aligned} Z(E) dE &= \frac{1}{8} \frac{4}{3} \pi [(R+dR)^3 - R^3] \\ &= \frac{1}{8} \frac{4}{3} \pi [\cancel{R^3} + \cancel{3R^2 dR} + 3R^2 dR + \cancel{3R dR^2} + \cancel{dR^3}] \\ &= \frac{1}{8} \frac{4}{3} \pi \cdot 3R^2 dR \end{aligned}$$

$$Z(E) dE = \frac{\pi}{2} R^2 dR \quad - (2)$$

Since $R^2 = \frac{E}{E_0}$

$$2R dR = \frac{1}{E_0} dE$$

$$R^2 dR = \frac{1}{2E_0} \sqrt{\frac{E}{E_0}} dE$$

$$R^2 dR = \frac{1}{2E_0^{3/2}} E^{1/2} dE$$

$$\begin{aligned} Z(E) dE &= \frac{\pi}{2} \frac{1}{2E_0^{3/2}} E^{1/2} dE \\ &= \frac{\pi}{4} \left(\frac{8mL^2}{h^2} \right)^{3/2} E^{1/2} dE \end{aligned}$$

$$Z(E) dE = \frac{\pi}{4} \frac{8 \times 2 \times 2^{1/2} m^{3/2} L^3}{h^3} E^{1/2} dE$$

$$Z(E) dE = \frac{\pi}{4} \left(\frac{8m}{h^2} \right)^{3/2} L^3 E^{1/2} dE$$

Now from the Pauli's exclusion principle, each state permits two electrons, so that the number of energy level actually available are

$$Z(E) dE = 2 \frac{\pi}{4} \left(\frac{8m}{h^2} \right)^{3/2} L^3 E^{1/2} dE$$

Hence the number of energy states presents in a unit volume

$$g(E) dE = \frac{\pi}{2} \left(\frac{8m}{h^2} \right)^{3/2} E^{1/2} dE$$

Thus the number of electrons $N(E) dE$ in a given energy interval dE can be found by multiplying the number of energy states per unit volume multiplied by Fermi distribution function

$$N(E) dE = \frac{\pi}{2} \left(\frac{8m}{h^2} \right)^{3/2} \frac{E^{1/2}}{\left(e^{\frac{E-E_F}{k_B T}} + 1 \right)} dE$$